

Harvard University
Physics 141 Pset 2

Due September 26, 2025

1. (a) Just prior to dividing, a bacterium contains 30 copies of the fluorescent protein GFP. Given that each protein has a $1/2$ chance of being included in each cell, regardless of the other proteins, what is the probability that we get exactly 15 copies of GFP in each daughter cell? How probable is it that one side gets fewer than 5 copies of GFP?
(b) If 7 quanta are absorbed by 500 rod cells in the visual field, what is the probability that two quanta will be absorbed by a single rod?
2. One way to model photons hitting the opsins in an eye is through a Poisson point process. In a time-homogeneous Poisson point process, each constant interval of time has the same probability distribution of the number of events that will happen. For example, , we can model the distribution of photons hitting an opsin molecule as

$$P(\text{total number of photons at time } t \text{ seconds} = n) = \frac{(at)^n}{n!} e^{-at}$$

for some constant a . First show that the distribution is normalized, and then find the mean and variance of the number of photons in the first second.

Hint: you won't be able to use an integral here. Try using a sum - does it look like any infinite series you can recognize?

3. Now, calculate the skewness

$$\left\langle \left(\frac{x - \langle x \rangle}{\sigma} \right)^3 \right\rangle$$

where x is total number of photons at 1 second. As a grows, what happens to this value?

4. In lecture, we discussed how optimizing the signal to noise ratio in a rod cell gives the counterintuitive result that 30% of photons should pass through undetected. Recall that the signal to noise ratio is:

$$\frac{(N_{\text{flash}}(1 - e^{-\sigma CL}))^2}{CALr_{\text{dark}}\tau}$$

where L is the length of the cell, C is the concentration of rhodopsin, σ is the absorption cross section of rhodopsin, A is the cross-sectional area of the rod cell, n_{flash} is the number of incident photons, r_{dark} is the rate of spontaneous thermal isomerizations, and τ is the integration time.

- (a) Optimize the signal to noise ratio and show that it is maximized when $L \approx \frac{1.26}{\sigma C}$. Feel free to use Wolfram Alpha to approximate transcendental equations.
 - (b) Cats, dogs, and many other animals have a reflective layer of tissue in their eye known as the *tapetum lucidum*, which gives them superior night vision to humans. Assuming that the *tapetum lucidum* acts as a perfect mirror at the end of the rod cell, repeat the signal to noise optimization. What percent of photons would pass undetected through the rod cell of a cat? Discuss the result. How does the *tapetum lucidum* improve the signal to noise ratio?
5. Read and understand figures 6 and 7 and table 5 in [Hecht, Schlaer, and Pirenne's paper](#).
- (a) What distributions did they plot for figure 6? Give $n = 1$ through $n = 3$.
 - (b) Recreate the plots for figure seven, along with the data. What is the R^2 for each graph? You may have to tune the x -shift accordingly; it may help to use the built in regressor on [desmos.com](#).
 - (c) What happens if the observers are pooled together? Find the optimal n and the resulting R^2 . What conclusion would they draw for the number of critical events in the retina required to produce a visual effect?

Feel free to attach your Desmos link (or alternative code/computational software).

6. **Stirling's Approximation (from Physical Biology of the Cell):** The Stirling approximation is useful in probability and combinatorics and is used to find an alternative approximation of $n!$.
- (a) First, show that $n! = \int_0^\infty x^n e^{-x} dx$. Use induction.
 - (b) Instead of analyzing the integrand $x^n e^{-x}$ directly, we will analyze the natural logarithm $f(x) = \ln(x^n e^{-x})$. Find where this function reaches its maximum.
 - (c) Expand $f(x)$ around some arbitrary positive value a to second order using a Taylor series. Exponentiate the result to get an approximation for $x^n e^{-x}$ around a .
 - (d) Plug in the maximum from part (b) to the value a in part c and simplify to get

$$n! \approx n^n e^{-n} \int_{-\infty}^{\infty} e^{-x^2/2} dx$$

Explain why we can modify the bounds to $\pm\infty$ and why we plug in the maximum.

- (e) Evaluate the integral and show that

$$n! \approx n^n e^{-n} (2\pi n)^{1/2}$$

7. In this problem, we will study and simulate a photon-counting detector. In a photon-counting detector, the number of photons detected in a time window (denoted N) is random and can be described via a Poisson distribution.

$$P(N = n) = e^{-\bar{n}} \frac{\bar{n}^n}{n!}$$

Where $n = 0, 1, 2, \dots$ and where $\bar{n}$ denotes the mean photon count. Conditioned on the number of photons detected being equal to n (i.e. $N = n$), the current you measure via the detector depends on two terms: (1) a background term B that is Gaussian distributed with mean 0 and variance σ_0^2 . This background term arises because of small fluctuations in electronics. (2) A photon-detection term S that consists of $n = N$ independent single-photon contributions that are Gaussian with mean i_1 and variance σ_1^2 (each photon results in a pulse in current i , this pulse is Gaussian distributed). With this information, we can show analytically that the expected distribution of current i is given by

$$P(i) = \sum_{n=0}^{\infty} P(i|n) e^{-\bar{n}} \frac{\bar{n}^n}{n!}$$

Where

$$P(i|n) = \frac{1}{\sqrt{2\pi(\sigma_0^2 + n\sigma_1^2)}} \exp\left(-\frac{(i - ni_1)^2}{2(\sigma_0^2 + n\sigma_1^2)}\right)$$

- (a) Explain in a few sentences why we can use a Poisson distribution to model the number of photons N that hit the detector? Why can we assume that the background current is Gaussian?
 - (b) We will now simulate the photon detection process and see how well it matches our analytical equation for $P(i)$. Take $i_1 = 1$, $\sigma_0 = 0.1$, $\sigma_1 = 0.2$, and $\bar{n} = 1$. In a programming language of your choice, generate a normalized histogram of the currents i . To do so, perform 10000 trials of the following: For each trial, first draw N from $Poisson(\bar{n})$. Then, draw a background B from $\mathcal{N}(0, \sigma_0^2)$ (this denotes a Gaussian with mean 0 and variance σ_0^2). Draw S by taking the sum of $n = N$ numbers sampled from $\mathcal{N}(i_1, \sigma_1^2)$. The current is $i = S + B$. Provide a normalized (area = 1) histogram of these simulated currents i for the 10000 trials.
 - (c) On the previous plot, also display the analytical equation for $P(i)$ for comparison. Since the sum in $P(i)$ is infinite, only evaluate the sum for $n_{max} = 10$ terms.
 - (d) Explore the impact of changing the bin size of your histogram and number of trials of your simulation on your results. How do binning and changing the number of trials affect the similarity between your simulation and the analytical result?
8. How long did this problem set take you to complete?